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Treat retaining pet toy

Abstract

A pet toy and method for occupying a pet. In one embodiment, the pet toy comprises a cylindrical main body member including a top end having a top wall, a bottom end opposite the top end that is open, and an annular side wall between the top end and the bottom end. The pet toy further comprises a core member configured to insert into the bottom end of the cylindrical main body member, and having a treat container configured to contain a pet treat. The top wall of the cylindrical main body member includes a treat access hole facing toward the treat container when the core member is inserted into the cylindrical main body member, and further includes one or more vents that serve multiple functions including relieving suction forces during use and redirecting saliva into the pet toy to prevent unwanted mess.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to provisional application 63/268,400, filed on Feb. 23, 2022, entitled “Structurally Reinforced Treat Housing For Occupying Dogs” and herein incorporated by reference.

FIELD

(1) The disclosure relates to the field of accessories for animals or pets, and in particular, to a pet toy for pets.

BACKGROUND

(2) Domestic animals are capable of providing a great deal of emotional fulfillment and support for their caretakers. However, many domestic animals (referred to generally herein as pets) demand a substantial amount of attention from their caretaker, especially while that caretaker is physically present. Thus, it remains difficult for a caretaker to balance other activities with the demands of a pet. This issue has only been exacerbated by recent trends increasing the prevalence of remote work and after-hours work in modern society. During work, a caretaker may be required to spend time attending to other tasks than mollifying a pet. However, the mere presence of the caretaker may increase the very desire and demand of the pet for attention.

(3) Hence, those who care for pets continue to desire devices for entertaining those pets while also carving out additional free time.

SUMMARY

(4) Embodiments described herein provide enhanced devices for retaining edible articles in a manner that is accessible, but also time-consuming, for a pet (e.g., a dog) to acquire. Devices as described herein are generally referred to as pet toys, and the edible articles are generally referred to as treats or pet treats. The pet toys described herein are re-usably separable by a caretaker into individual components or members. This enables a treat to be loaded into a treat container of the pet toy while the individual members are separated. When the individual members are joined, the treat is trapped in the treat container, but a pet has access to the pet treat through a treat access hole. Thus, the pet may interact with the pet treat via smell and taste, but is incapable of instantly devouring the treat as it is trapped in the treat container.

(5) Pet toys and methods described herein provide a benefit by drawing the attention of a pet for an extended period of time as the pet attempts to consume a pet treat. This in turn grants caretakers additional free time to pursue their own goals while the pet is occupied.

- (6) In an embodiment, a pet toy comprises a cylindrical main body member comprising a top end having a top wall, a bottom end opposite the top end that is open, and an annular side wall between the top end and the bottom end. The pet toy further comprises a core member configured to insert into the bottom end of the cylindrical main body member, and having a treat container configured to contain a pet treat. The top wall of the cylindrical main body member includes a treat access hole facing toward the treat container when the core member is inserted into the cylindrical main body member. The top wall of the cylindrical main body member further includes one or more vents.
- (7) In an embodiment, the treat access hole is radially centered on the top wall of the cylindrical main body member, and a plurality of the vents are disposed radially outward from the treat access hole.
- (8) In an embodiment, the annular side wall projects axially outward in relation to the top wall where the annular side wall and the top wall intersect to form an annular rim.
- (9) In an embodiment, the annular rim slopes axially inward toward the vents.
- (10) In an embodiment, the cylindrical main body member further comprises a plurality of grooves disposed axially along the annular side wall.
- (11) In an embodiment, the core member comprises a top end that is open, a bottom end opposite the top end that is closed at a base, and an annular side wall between the top end and the bottom end of the core member. An inner side surface and an inner bottom surface of the core member forms the treat container.
- (12) In an embodiment, the treat container is generally bowl-shaped.
- (13) In an embodiment, the treat container is configured to contain a liquid.
- (14) In an embodiment, contact between the cylindrical main body member and the core member forms a liquid seal between the cylindrical main body member and the core member when the core member is inserted into the cylindrical main body member.
- (15) In an embodiment, the base comprises an annular flange that abuts the bottom end of the cylindrical main body member when the core member is inserted into the cylindrical main body member.
- (16) In an embodiment, the top wall of the cylindrical main body member is rounded axially outward, and the base of the core member is rounded axially outward.
- (17) In an embodiment, a center of gravity of the pet toy is disposed closer to the base than the top end of the cylindrical main body member to bias the top wall of the cylindrical main body member upward.
- (18) In an embodiment, the core member comprises a grip disposed on the base configured to be grasped by a human hand.
- (19) In an embodiment, a kit comprises as pet toy as described above, and the pet treat having a dimension larger than a dimension of the treat access hole so that the top wall of the cylindrical main body member traps the pet treat in the treat container of the core member.
- (20) In an embodiment, a kit comprises the pet toy as described above, and a mold configured to shape the pet treat having a dimension larger than a dimension of the treat access hole so that the top wall of the cylindrical main body member traps the pet treat in the treat container of the core member.
- (21) In an embodiment, a pet toy comprises an upper member and a lower member that are detachable. The upper member defines a hollow cylinder, and comprises a top end having a top wall, a bottom end opposite the top end that is open, and an annular side wall between the top end and the bottom end. The lower member is configured to screw into the bottom end of the upper member, and includes a liquid-impermeable treat container configured to contain a pet treat. The top wall of the upper member is configured to trap the pet treat in the treat container when the lower member is screwed into the upper member, and includes a treat access hole and a plurality of vents.
- (22) In an embodiment, the treat access hole is radially centered on the top wall of the upper

member, and the vents are disposed radially outward from the treat access hole.

(23) In an embodiment, the annular side wall projects axially outward in relation to the top wall where the annular side wall and the top wall intersect to form an annular rim, and the annular rim slopes axially inward toward the vents.

(24) In an embodiment, contact between the upper member and the lower member forms a liquid seal between the upper member and the lower member when the lower member is screwed into the upper member.

(25) In an embodiment, a method of occupying a pet comprises obtaining a pet toy as described above, obtaining a pet treat, loading the pet treat in the treat container of the core member with the core member detached from the cylindrical main body member, and inserting the core member into the bottom end of the cylindrical main body member so that the treat access hole is facing toward the treat container. The top wall of the cylindrical main body member traps the pet treat in the treat container while providing access to the pet treat through the treat access hole.

(26) Other illustrative embodiments may be described below. The features, functions, and advantages that have been discussed can be achieved independently in various embodiments or may be combined in yet other embodiments further details of which can be seen with reference to the following description and drawings.

Description

DESCRIPTION OF THE DRAWINGS

(1) Some embodiments of the present disclosure are now described, by way of example only, and with reference to the accompanying drawings. The same reference number represents the same element or the same type of element on all drawings.

(2) FIG. 1 is a perspective view of a pet toy configured to retain a pet treat in an illustrative embodiment.

(3) FIG. 2 is a top view of the pet toy in an illustrative embodiment.

(4) FIG. 3 is a side view of the pet toy in an illustrative embodiment.

(5) FIG. 4 is a section cut view of the pet toy in an illustrative embodiment.

(6) FIG. 5 is a bottom view of the pet toy in an illustrative embodiment.

(7) FIG. 6 is a perspective view of a lower member of the pet toy in an illustrative embodiment.

(8) FIG. 7 is a side view of the lower member of the pet toy in an illustrative embodiment.

(9) FIG. 8 is a section cut view of the lower member of the pet toy in an illustrative embodiment.

(10) FIG. 9 is a perspective view of an upper member of the pet toy in an illustrative embodiment.

(11) FIG. 10 is a side view of the upper member of the pet toy in an illustrative embodiment.

(12) FIG. 11 is a section cut view of the upper member of the pet toy in an illustrative embodiment.

(13) FIG. 12 is a perspective view of a pet toy in another illustrative embodiment.

(14) FIG. 13 is a section cut view of a pet toy in another illustrative embodiment.

(15) FIG. 14 is a perspective view of a pet toy in another illustrative embodiment.

(16) FIG. 15 is a section cut view of a pet toy in another illustrative embodiment.

(17) FIG. 16 is a block diagram depicting a relationship of dimensions between a treat access hole, a pet treat, and a treat container in an illustrative embodiment.

(18) FIG. 17 is a perspective view of a mold for pet treats in an illustrative embodiment.

(19) FIG. 18 is a top view of the mold in an illustrative embodiment.

(20) FIG. 19 is a side view of the mold in an illustrative embodiment.

(21) FIG. 20 is a flow chart illustrating a method of occupying a pet in an illustrative embodiment.

DESCRIPTION

(22) The figures and the following description depict specific illustrative embodiments of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise various

arrangements that, although not explicitly described or shown herein, embody the principles of the disclosure and are included within the scope of the disclosure. Furthermore, any examples described herein are intended to aid in understanding the principles of the disclosure, and are to be construed as being without limitation to such specifically recited examples and conditions. As a result, the disclosure is not limited to the specific embodiments or examples described below, but by the claims and their equivalents.

(23) FIG. 1 is a perspective view of a pet toy **100** configured to retain a pet treat in an illustrative embodiment. Pet toy **100** is an apparatus, object, or device designed for a pet, such as a dog. Pet toy **100** is enhanced to removably store a pet treat in a manner that prevents, hinders, or obstructs a pet from fully retrieving or extracting the pet treat from pet toy **100**. At the same time, pet toy **100** grants limited access for a pet to reach the pet treat (e.g., by the pet extending its tongue into pet toy **100**) via a treat access hole **110**. This keeps the pet entertained for a longer period of time than would be granted by a treat dispenser or a simple toy. For example, the pet may use its tongue to abrade the pet treat trapped in the pet toy **100**, and consume the pet treat over a long period of time, such as thirty to forty minutes. Pet toy **100** may be scaled to any suitable size for a pet. For example, pet toy **100** intended for use by dogs may have a circumference of about 3π inches, 3.5π inches, etc.

(24) In this embodiment, pet toy **100** comprises an upper member **102** and a lower member **150** that are detachable or separable into individual components. Upper member **102** and lower member **150** are configured to be repeatedly detached from, and attached to, each other in order to assemble pet toy **100**. This enables pet toy **100** to be opened or disassembled for replacement of a pet treat and/or for cleaning. Pet toy **100** may further be closed or assembled to prevent, hinder, or obstruct pet treats from being quickly retrieved or extracted from pet toy **100** by a pet. Upper member **102** and lower member **150** are arranged about a center axis **114** proceeding in the Y direction. In this embodiment, center axis **114** occupies a center of cross-sections of pet toy **100** along the XZ plane.

(25) Upper member **102** (also referred to as a cylindrical main body member) comprises a main body member of pet toy **100**, and is generally cylindrical in shape. Upper member **102** is a monolithic body that includes a top end **103**, a bottom end **107**, and an annular side wall **105** between the top end **103** and the bottom end **107**. Annular side wall **105** provides physical support and strength to pet toy **100** that prevents pet toy **100** from being damaged, crushed, or separated by a pet. For example, annular side wall **105** may be dimensioned for sufficient structural strength to resist crushing by the jaws of a pet. In an embodiment, annular side wall **105** is made from a pliable or flexible material **109**, such as a rubber (e.g., a silicone rubber), that resists damage and permanent deformation. Flexible material **109** may also be soft enough to ensure that the teeth of a pet are not damaged by interaction with the flexible material **109**.

(26) An outer surface **104** of annular side wall **105** may exhibit a curvature, such as a convex curvature, between top end **103** and bottom end **107**. In other words, outer surface **104** of annular side wall **105** may curve radially outward between top end **103** and bottom end **107** around the circumference of annular side wall **105**. A curvature of outer surface **104** facilitates rolling, bouncing, or other dynamic action when pet toy **100** is thrown, nudged, kicked, or butted.

(27) Bottom end **107** of upper member **102** is open to receive lower member **150** as will be described in more detail below. Top end **103** of upper member **102** is partially closed to define a treat interaction portion **111**. Treat interaction portion **111** is the portion of pet toy **100** that serves dual purposes of trapping a pet treat in pet toy **100**, and allows a pet to interact with a pet treat trapped or contained in pet toy **100**. Treat interaction portion **111** is formed by a top wall **113** on top end **103**. In one embodiment, top wall **113**/treat interaction portion **111** includes a treat access hole **110**. Treat access hole **110** is an opening, orifice, or void in top wall **113** that provides a pet access to a pet treat. Treat access hole **110** is dimensioned to allow passage or penetration of a pet tongue into an interior of pet toy **100**. Thus, treat access hole **110** is wide enough for a dog tongue or another type of pet. Treat access hole **110** is circular in this embodiment, but may have other shapes

in other embodiments, such as an oval. Also, a single treat access hole **110** is shown in the embodiment of FIG. **1**, but more than one treat access hole **110** may be implemented on top wall **113** in other embodiments.

(28) Top wall **113**/treat interaction portion **111** further includes one or more vents **112**. A vent **112** is an opening, slot, orifice, or void in top wall **113** disposed on the same top end **103** as treat access hole **110** (i.e., distinct from treat access hole **110**). A size or dimension of vents **112** are smaller than a size or dimension of treat access hole **110**. Vents **112** may be generally rectangular in shape (as opposed to a circular shape of treat access hole **110**), although other shapes are considered herein. The size and/or shape of vents **112** may be selected to prevent penetration of a pet tongue into an interior of pet toy **100** through vents **112**. In one embodiment, treat access hole **110** may be disposed toward a center of top wall **113** (i.e., centered on center axis **114**), with vents **112** disposed radially outward from treat access hole **110**. Vents **112** are configured to relieve back pressure in pet toy **100** when treat access hole **110** is plugged by the tongue of a pet. Vents **112** therefore provide a technical benefit by preventing a suction lock from occurring when a pet inserts its tongue into treat access hole **110** to reach a pet treat within an interior of pet toy **100**. That is, vents **112** enable airflow into pet toy **100** from the same end as treat access hole **110** even when treat access hole **110** is entirely blocked by a tongue, which prevents the tongue from becoming trapped within pet toy **100** by a pressure differential. Vents **112** also provide a function of saliva or drool capture, which is discussed in more detail below.

(29) Although the structure of the top wall **113** of upper member **102** may vary, top wall **113** may include buttresses **106** (also referred to herein as “supports”) that project radially inward from annular side wall **105** towards the center axis **114**, and project axially outward in the Y direction in relation to annular side wall **105**. Top wall **113** may further include a ring **108**, which defines treat access hole **110** from an exterior **190** of pet toy **100** to an interior portion of pet toy **100**. In further embodiments, buttresses **106** may vary in thickness, and may comprise supports that are thick enough to appear as a continuous ring interrupted by vents **112**.

(30) In an embodiment, annular side wall **105** may project axially outward in relation to the top wall **113** where annular side wall **105** and the top wall **113** intersect to form an annular rim **116**. Annular rim **116** slopes or curves axially inward toward vents **112** of top wall **113**. Thus, annular rim **116** defines an annular trough **117** along top wall **113** that facilitates the redirection of saliva or drool on top wall **113** toward vents **112**, as discussed with regard to FIG. **4** below. In addition to preventing a suction lock, vents **112** also function as an inlet for saliva or drool into an interior portion of pet toy **100**.

(31) Lower member **150** (also referred to as a core member or plug member) is generally cylindrical in shape. Lower member **150** comprises a top end (not visible in FIG. **1**), a bottom end **151** opposite the top end that is closed at a base **152**, and an annular side wall (not visible in FIG. **1**) between the top end and the bottom end **151**. Base **152** of lower member **150** represents the bottom of pet toy **100** that rests on a floor when pet toy **100** is oriented upright with top end **103** facing upward. As will be described in more detail below, an interior portion of lower member **150** forms a treat container. Lower member **150** may be made from the flexible material **109** or a rigid material **154** (e.g., a plastic). In an embodiment, base **152** is curved or rounded axially outward or generally has a spherical or rounded outline, which enables pet toy **100** to roll when dropped or thrown. This increases the entertainment value of pet toy **100**, which may be used by the pet in a similar manner to a ball.

(32) FIG. **2** is a top view of pet toy **100** in an illustrative embodiment. FIG. **2** illustrates top wall **113** and treat interaction portion **111** having an arrangement of treat access hole **110** and vents **112**. In one embodiment, vents **112** may be spaced circumferentially around top wall **113** and separated by buttresses **106**. Although five vents **112** are shown in FIG. **2**, a different number of vents **112** may be implemented in other embodiments. Treat access hole **110** may be disposed generally toward a center **240** of top wall **113**, and may curve or slope axially inward toward center **240**.

(33) Furthermore, FIG. 2 illustrates that an outer surface **104** of annular side wall **105** may include one or more grooves **220** disposed radially inward from outer surface **104**. Grooves **220** extend or are disposed generally vertical or axial on outer surface **104** along the Y-axis, and may be referred to as vertical grooves. However, in other embodiments, grooves **220** may extend at an angle to the Y-axis. Grooves **220** may be circumferentially arranged about outer surface **104** as shown in FIG. 2. Grooves **220** provide multiple technical benefits. First, the addition of grooves **220** causes motion of pet toy **100** to become more erratic when thrown or rolled, which may cause pet toy **100** to appear more enticing for a pet to pursue. Second, the use of grooves **220** facilitates gripping of pet toy **100** by a human. This makes the process of separating and re-joining upper member **102** and lower member **150** more ergonomic, particularly in embodiments where the upper member **102** and the lower member **150** are mated. Third, the existence of grooves **220** facilitates the cleaning of teeth of a pet when pet toy **100** is chewed or gnawed by the pet. Although outer surface **104** is described as having one or more grooves **220**, the outer surface **104** may have different types of surface features, such as one or more ridges **210** that project radially outward from outer surface **104**, a plurality of bumps arranged on outer surface **104** (see, for example, FIG. 12), etc. Alternatively, outer surface **104** may be generally smooth in other embodiments (see, for example, FIG. 14).

(34) FIG. 3 is a side view of pet toy **100** in an illustrative embodiment. As is evident, top wall **113** of upper member **102** may be curved or rounded axially outward, and base **152** of lower member **150** may be curved or rounded axially outward in the opposite direction. Likewise, outer surface **104** of annular side wall **105** of upper member **102** may be curved or rounded radially outward. FIG. 3 further illustrates a geometry and distribution of grooves **220** around outer surface **104** of upper member **102**. FIG. 3 also provides further detail of lower member **150**, and illustrates a grip **300** disposed on base **152** of lower member **150**. Grip **300** is a surface feature on base **152** configured to be grasped by a human hand, such as a handle. Grip **300** may be used for coupling and de-coupling the lower member **150** to/from the upper member **102**. For example, by holding the upper member **102** via grooves **220** in one hand and holding the grip **300** of lower member **150** in the other hand, a caretaker may ergonomically apply torque that joins/separates the lower member **150** and the upper member **102**.

(35) FIG. 4 is a section cut view of pet toy **100** in an illustrative embodiment, and corresponds with view arrows 4-4 of FIG. 2. In an embodiment, where annular side wall **105** and top wall **113** of upper member **102** intersect, annular side wall **105** may extend or project axially outward from top wall **113** to form annular rim **116**. Annular rim **116** may slope or curve radially inward towards the vents **112**, which provides a technical benefit by capturing and re-directing saliva that is flowing along top wall **113**. Annular rim **116** redirects the saliva into an interior portion **412** of pet toy **100** and into a treat container **410** that contains the pet treat, instead of onto the outer surface **104** of upper member **102**. Treat container **410** is configured to hold or contain liquid so the saliva is contained within pet toy **100** resulting in less of a mess. This also prevents the outer surface **104** from becoming overly slick, which may interfere with the disassembly of pet toy **100**, as well as the handling of pet toy **100** when used as a ball or toy.

(36) FIG. 4 depicts the annular side wall **452** of lower member **150** in an illustrative embodiment. The annular side wall **452** of lower member **150** is generally cylindrical and dimensioned to mate (i.e., fit into) with annular side wall **105** of upper member **102**. In one embodiment, annular side wall **452** may include threads **434** on an outer side surface, while annular side wall **105** includes threads **432** on an inner side surface. In this embodiment, threads **432** and **434** are dimensioned to mate when lower member **150** is screwed (i.e., united by means of a twisting motion) into upper member **102**. However, in other embodiments, other mating features, such as magnets, interlocking teeth, a zipper, a button, interlocking ridges, etc., may be utilized to securely hold lower member **150** when lower member **150** is inserted in upper member **102** to allow for repeated assembly and disassembly of the lower member **150** to/from the upper member **102**.

(37) In one embodiment, treat container **410** is defined by an interior portion of lower member **150**. A treat container **410** is an internal cavity or compartment within lower member **150** that is configured to contain a pet treat. Treat container **410** may be generally bowl-shaped **445** that is open at its top, and has a continuous inner side surface **478** and a continuous inner bottom surface **486**. Treat container **410** is configured to hold or contain a liquid (i.e., liquid impermeable), meaning that treat container **410** is devoid of holes on inner side surface **478** or inner bottom surface **486** so that liquid cannot flow out of treat container **410** axially through inner bottom surface **486** or radially out of inner side surface **478**. When assembled as in FIG. 4, treat access hole **110** is facing toward treat container **410**. Treat access hole **110** may be considered as being in axial alignment with treat container **410** when pet toy **100** is assembled, where treat access hole **110** is substantially aligned axially with treat container **410**. One technical benefit is a frozen pet treat or the like may be contained in treat container **410**, and liquid melting from the frozen pet treat cannot escape treat container **410** other than through top wall **113**. Another technical benefit is saliva captured in treat container **410** cannot escape other than through top wall **113**.

(38) In one embodiment, treat container **410** may include a central depression **450** at inner bottom surface **486**. Treat container **410** may further include one or more peripheral depressions **454** arranged radially outward from central depression **450** at inner bottom surface **486**. Central depression **450** and/or peripheral depression **454** function to trap at least a portion of a pet treat to extend playtime by the pet. In further embodiments, the number, depth, arrangement, and even existence of the various depressions of treat container **410** may be varied as a matter of design choice.

(39) Additional features of pet toy **100** may facilitate sealing of upper member **102** and lower member **150** so that liquid cannot escape or leak between these members. In general, a liquid seal **490** may be formed between upper member **102** and lower member **150** due to contact between upper member **102** and lower member **150** when lower member **150** is inserted into upper member **102**. For example, with lower member **150** tightened or fully inserted/fully screwed into upper member **102**, a liquid-tight or liquid-proof seal **490** may be formed between upper member **102** and lower member **150**. For example, an annular lip **460** of upper member **102** may abut an upper surface **462** of annular side wall **452** of lower member **150** when upper member **102** and lower member **150** are joined. Annular lip **460** may project radially inward with respect to threads **432** to abut upper surface **462** of annular side wall **452**. Additionally or alternatively, a liquid seal **490** may be formed via engagement of threads **432** with threads **434**, when threads are implemented as mating features. A liquid seal **490** provides a technical benefit by preventing liquid from penetrating or leaking between lower member **150** and upper member **102**. Further, preventing penetration of liquid between threads **432** and threads **434** ensures that upper member **102** and lower member **150** do not loosen with respect to each other, and also reduces the need for washing the threads **432** and **434** between uses.

(40) An additional feature illustrated by FIG. 4 is a flange **470** at the base **152** of lower member **150**. Flange **470** projects radially outward with respect to annular side wall **452** to abut the bottom end **107** of upper member **102**. The outermost portion of flange **470** is generally flush with outer surface **104** of annular side wall **105**. Thus, the outermost portion of flange **470** extends a contour **472** of the outer surface **104**, such that the contour **472** continues substantially smoothly into a contour **474** at the base **152** of lower member **150**. This provides a technical benefit by facilitating rolling of pet toy **100** when pet toy **100** is perturbed.

(41) FIG. 4 further illustrates that pet toy **100** may exhibit a biased center of gravity. That is, a center of gravity **482** of pet toy **100** might not be disposed at a geometric center of pet toy **100**. Rather, the center of gravity **482** may be disposed closer to a lower surface **484** of the base **152** than top end **103** of upper member **102**, such as within volume **480**. This causes pet toy **100** to roll erratically, which may be enticing to a pet. Furthermore, offsetting the center of gravity **482** causes pet toy **100** to bias the top wall **113**/treat interaction portion **111** (i.e., treat access hole **110**) upward

when pet toy **100** is rolled, and after pet toy **100** has finished moving. When biased upward, the bottom of pet toy **100** rests on the floor with top end **103** facing upward, which helps to ensure that the treat access hole **110** remains accessible to the pet after pet toy **100** has been perturbed; enticing further interaction. Upward biasing of pet toy **100** also prevents liquid (e.g., drool, saliva, melted pet treat, etc.) from escaping from treat container **410** if left unused.

(42) FIG. 5 is a bottom view of pet toy **100** in an illustrative embodiment. As shown in FIG. 5, lower member **150** is inserted (e.g., screwed) into annular side wall **105** of upper member **102**.

When assembled in this manner, lower member **150** and upper member **102** are concentric cylindrical members. In one embodiment, grip **300** is dimensioned for ergonomic twisting by a human hand, and hence forms a narrow arch, which enables lower member **150** to be turned like a dial when screwed into upper member **102**.

(43) With the prior FIGS. 1-5 depicting an assembled pet toy **100** for retaining a pet treat, FIGS. 6-8 further depict lower member **150** in an illustrative embodiment.

(44) FIG. 6 is a perspective view of lower member **150** of pet toy **100** in an illustrative embodiment. Lower member **150** is a monolithic body that includes top end **602**, a bottom end **151**, and an annular side wall **452** between the top end **602** and the bottom end **151**. In this view, treat container **410** is defined by inner side surface **478** and inner bottom surface **486** (not visible in FIG. 6) of lower member **150**. Inner side surface **478** is generally cylindrical in this embodiment, and is continuous (i.e., not including holes). Outer side surface **610** of annular side wall **452** is generally a cylindrical shape between base **152** and a top end **602** of lower member **150**. In one embodiment, outer side surface **610** of annular side wall **452** may include threads **434** as a mating feature. The bottom end **151** of lower member **150** (at base **152**) is closed, while top end **602** is generally open to form treat container **410**. Treat container **410** extends axially downward from upper surface **462** of annular side wall **452** to inner bottom surface **486** (see FIG. 4).

(45) FIG. 7 is a side view of lower member **150** in an illustrative embodiment, providing additional perspective on the features of lower member **150**. Specifically, FIG. 7 provides further detail depicting flange **470**. Flange **470** may project radially outward from outer side surface **610** of lower member **150**. The upper surface **772** of flange **470** may curve or angle downward (in the Y direction) as flange **470** projects radially outward. Upper surface **772** may be configured to abut a bottom end **107** (see FIG. 1) of upper member **102** when upper member **102** and lower member **150** are assembled. FIG. 7 also illustrates that base **152** of lower member **150** may be curved or rounded axially outward.

(46) FIG. 8 is a section cut view of lower member **150** in an illustrative embodiment, and corresponds with view arrows 4-4 of FIG. 2. As described above, inner side surface **478** extends axially downward from upper surface **462** of annular side wall **452** to inner bottom surface **486** to form treat container **410**. Inner bottom surface **486** of treat container **410** may include central depression **450**, and/or one or more peripheral depressions **454** arranged radially outward from central depression **450**.

(47) FIGS. 9-11 depict upper member **102** in an illustrative embodiment. As in FIG. 9, upper member **102** is generally a cylindrical shape between top end **103** and bottom end **107** (note that upper member **102** is flipped upside down in FIG. 9) that defines a hollow cylinder. The bottom end **107** of upper member **102** is generally open so that an interior of upper member **102** is generally hollow. In one embodiment, an inner side surface **914** of upper member **102** may include threads **432** as a mating feature.

(48) In FIG. 10, upper member **102** is illustrated from a side view. FIG. 10 illustrates that top wall **113** of upper member **102** may be curved or rounded axially outward. Also, an annular rim **116** may project axially outward from a lower portion of vents **112** to funnel any liquid on top wall **113** toward vents **112** instead of along outer surface **104**.

(49) FIG. 11 is a section cut view of upper member **102** that corresponds with view arrows 4-4 of FIG. 2. FIG. 11 illustrates an interior portion **1110** of the upper member **102** that is generally

hollow, and an inner side surface **914** of annular side wall **105** facing radially inward toward the interior portion **1110**. In FIG. **11**, vents **112** may include an outer radial surface **1102** that slopes or curves radially inward from annular rim **116** toward interior portion **1110**. Annular rim **116** may project axially outward further than a lower portion **1105** and a radially outermost portion **1104** of vent **112**. Thus, annular rim **116** obstructs saliva from traveling down outer surface **104** of upper member **102**, and instead funnels the saliva or other liquid down the outer radial surface **1102** of vents **112**.

(50) FIG. **12** is a perspective view of a pet toy **100** in another illustrative embodiment. As above, pet toy **100** comprises an upper member **102** and a lower member **150** that are detachable or separable into individual components. Upper member **102** includes a top end **103**, a bottom end **107**, and an annular side wall **105** between the top end **103** and the bottom end **107**. Bottom end **107** of upper member **102** is open to receive lower member **150** as will be described in more detail below. Top end **103** of upper member **102** is partially closed to define a treat interaction portion **111**. Treat interaction portion **111** is formed by a top wall **113** on top end **103**. In one embodiment, top wall **113**/treat interaction portion **111** includes a treat access hole **110**, and one or more vents **112**.

(51) In an embodiment, annular side wall **105** may project axially outward in relation to the top wall **113** where annular side wall **105** and the top wall **113** intersect to form an annular rim **116**. Annular rim **116** slopes or curves axially inward toward vents **112** of top wall **113**. Thus, annular rim **116** facilitates the redirection of saliva or drool on top wall **113** toward vents **112**. In addition to preventing a suction lock, vents **112** also function as an inlet for saliva or drool into an interior portion of pet toy **100**.

(52) FIG. **12** further illustrates that annular side wall **105** may include one or more bumps **1212** that project radially outward from outer surface **104**. Bumps **1212** are surface features, as an alternative or addition to grooves **220** shown in FIG. **2**, that facilitate gripping of pet toy **100** by a human.

(53) Lower member **150** is generally cylindrical in shape. Lower member **150** comprises a top end (not visible in FIG. **1**), a bottom end **151** opposite the top end that is closed at a base **152**, and an annular side wall (not visible in FIG. **1**) between the top end and the bottom end **151**. Base **152** of lower member **150** represents the bottom of pet toy **100** that rests on a floor when pet toy **100** is oriented upright with top end **103** facing upward. As will be described in more detail below, an interior portion of lower member **150** forms a treat container. In an embodiment, base **152** is curved or rounded axially outward or generally has a spherical or rounded outline, which enables pet toy **100** to roll when dropped or thrown.

(54) FIG. **13** is a section cut view of pet toy **100** in an illustrative embodiment. In an embodiment, where annular side wall **105** and top wall **113** of upper member **102** intersect, annular side wall **105** may extend or project axially outward from top wall **113** to form annular rim **116**. Annular rim **116** may slope or curve radially inwards towards the vents **112**, which provides a technical benefit by capturing and re-directing saliva that is flowing along top wall **113**. Annular rim **116** redirects the saliva into an interior portion **412** of pet toy **100** and into a treat container **410** that contains the pet treat, instead of onto the outer surface **104** of upper member **102**. Treat container **410** is configured to hold or contain liquid so the saliva is contained within pet toy **100** resulting in less of a mess. This also prevents the outer surface **104** from becoming overly slick, which may interfere with the disassembly of pet toy **100**, as well as the handling of pet toy **100** when used as a ball or toy.

(55) FIG. **13** depicts the annular side wall **452** of lower member **150** in an illustrative embodiment. The annular side wall **452** of lower member **150** is generally cylindrical and dimensioned to mate (i.e., fit into) with annular side wall **105** of upper member **102**. In one embodiment, a clip mechanism **1302** may define the mating features of upper member **102** and lower member **150**. For clip mechanism **1302**, annular side wall **452** of lower member **150** may include one or more tab members **1304** that project radially outward from an outer side surface of annular side wall **452**. For example, a tab member **1304** may be an annular tab member disposed around the

circumference of annular side wall **452**. Alternatively, one or more tab members **1304** may be spaced around the circumference of annular side wall **452** (e.g., a pair of tab members **1304** disposed on radially-opposite sides of lower member **150**). Annular side wall **105** of upper member **102** may include one or more notches **1306** or indentations. In this embodiment, a tab member **1304** is dimensioned to mate with a notch **1306** when lower member **150** is pressed into upper member **102**. Upper member **102** may further include one or more detents **1310** configured to facilitate engagement/disengagement of a tab member **1304** with a notch **1306**.

(56) FIG. **14** is a perspective view of a pet toy **100** in another illustrative embodiment. As above, pet toy **100** comprises an upper member **102** and a lower member **150** that are detachable or separable into individual components. Upper member **102** includes a top end **103**, a bottom end **107**, and an annular side wall **105** between the top end **103** and the bottom end **107**. Bottom end **107** of upper member **102** is open to receive lower member **150** as will be described in more detail below. Top end **103** of upper member **102** is partially closed to define a treat interaction portion **111**. Treat interaction portion **111** is formed by a top wall **113** on top end **103**. In one embodiment, top wall **113**/treat interaction portion **111** includes a treat access hole **110**, and one or more vents **112**. In the embodiment of FIG. **14**, the outer surface **104** of annular side wall **105** may be generally smooth.

(57) Lower member **150** is generally cylindrical in shape. Lower member **150** comprises a top end (not visible in FIG. **1**), a bottom end **151** opposite the top end that is closed at a base **152**, and an annular side wall (not visible in FIG. **1**) between the top end and the bottom end **151**. As will be described in more detail below, an interior portion of lower member **150** forms a treat container. In an embodiment, base **152** may include a grip **300** dimensioned for a human hand, and the design of grip **300** may vary from the design illustrated in FIG. **14**, such as in FIG. **5**.

(58) FIG. **15** is a section cut view of pet toy **100** in an illustrative embodiment. FIG. **15** depicts the annular side wall **452** of lower member **150** in an illustrative embodiment. The annular side wall **452** of lower member **150** is generally cylindrical and dimensioned to mate (i.e., fit into) with annular side wall **105** of upper member **102**. In one embodiment, a clip mechanism **1302** may define the mating features of upper member **102** and lower member **150**. For clip mechanism **1302**, annular side wall **452** of lower member **150** may include one or more tab members **1304** that project radially outward from an outer side surface of annular side wall **452**. Annular side wall **105** of upper member **102** may include one or more notches **1306** or indentations. In this embodiment, a tab member **1304** is dimensioned to mate with a notch **1306** when lower member **150** is pressed into upper member **102**.

(59) One or more of the features disclosed above for pet toy **100** may be combined as desired.

(60) FIG. **16** is a block diagram **1600** depicting a relationship of dimensions between treat access hole **110**, a pet treat **1610**, and treat container **410** in an illustrative embodiment. Pet treat **1610** may comprise any suitable material that is readily consumable by a pet. Depending on the embodiment, pet treat **1610** may be non-spherical. For example, pet treat **1610** may have an ovoid, cylindrical, or non-uniform shape. In some embodiments, pet treat **1610** may comprise a paste, condensed food item, freeze dried item, frozen item, etc. Furthermore, in some embodiments, pet treat **1610** exhibits viscoelastic properties, and is capable of being slowly abraded or otherwise scraped away via interaction with a tongue of the pet.

(61) FIG. **16** illustrates that the dimensions of treat access hole **110**, treat container **410**, and pet treat **1610** may be carefully selected. That is, dimension **D1** of treat access hole **110**, comprising a widest or largest dimension of treat access hole **110**, is smaller than a dimension **D3** corresponding with a width of pet treat **1610** in the X direction. Similarly, dimension **D3** is smaller than dimension **D2**, which is a width of treat container **410**. These relationships in dimension provide a technical benefit by ensuring that the pet treat **1610** is capable of being held within treat container **410**, and yet restricting the pet treat **1610** from being dropped or pulled through treat access hole **110**. Thus, pet treat **1610** cannot be removed or dispensed via treat access hole **110**.

(62) With a discussion of variations of an apparatus for retaining pet treat described above, FIGS. **17-19** depict a mold for fabricating pet treats **1610** that may be stored within pet toy **100**. FIG. **17** is a perspective view of a mold **1700** for pet treats in an illustrative embodiment. For example, mold **1700** may be utilized to shape a food putty or paste, and/or for maintaining the shape of such a food putty or paste during freezing. In this embodiment, mold **1700** includes a body **1710**, which unites compartments **1720**. Each compartment **1720** defines a well **1730** or cup for receiving an edible material that may be shaped into a pet treat **1610**. The mold **1700** may be made from a material **1702**, such as a rubber or silicone rubber. The use of a flexible material provides a technical benefit by facilitating deformation of the mold **1700** to release pet treats **1610** from the wells **1730**.

(63) FIG. **18** is a top view of mold **1700** in an illustrative embodiment. In this embodiment, a base of each compartment **1720** may be stamped with an indicator referring to a type of edible material, such as indicator **1810** for vegetable material, an indicator **1820** for fish material, an indicator **1830** for beef material, and/or an indicator **1840** for poultry material.

(64) FIG. **19** is a side view of mold **1700** in an illustrative embodiment, and is provided to add further context. For example, FIG. **19** further illustrates stabilizers **1910** attached to each compartment **1720**. Stabilizers **1910** provide a technical benefit by helping to keep the mold **1700** level, which facilitates retention of food articles that are still being shaped (e.g., being frozen) at the mold **1700**.

(65) FIG. **20** is a flow chart illustrating a method **2000** of occupying a pet in an illustrative embodiment. The steps of the flow charts described herein are not all inclusive and may include other steps not shown, and the steps may be performed in an alternative order. Method **2000** includes obtaining a pet toy **100** as described above (step **2002**). Method **2000** further includes obtaining a pet treat **1610** (step **2004**). The pet treat **1610** has a larger size than the treat access hole **110** of pet toy **100**. Method **2000** further includes loading the pet treat **1610** in treat container **410** of lower member **150** with lower member **150** detached from upper member **102** (step **2006**). Method **2000** further includes inserting (e.g., screwing, twisting, pressing, etc.) lower member **150** into the bottom end **107** of upper member **102** so that treat access hole **110** is facing toward (or axially aligned with) treat container **410** (step **2008**). The top wall **113** of upper member **102** traps the pet treat **1610** in treat container **410** while providing/allowing access to the pet treat **1610** through treat access hole **110**.

(66) Although specific embodiments were described herein, the scope of the disclosure is not limited to those specific embodiments. The scope of the disclosure is defined by the following claims and any equivalents thereof

Claims

1. A pet toy comprising: a cylindrical main body member comprising: a top end having a top wall; a bottom end opposite the top end that is open; an annular side wall between the top end and the bottom end, comprising mating features on a side surface thereof; a treat access hole, disposed toward a center of the top wall and curving or sloping inward toward a center of the treat access hole; and one or more vents, disposed on the top wall, each vent including a lower surface that slopes or curves inward toward an interior portion of the pet toy to funnel liquid down the vents and into the interior portion, wherein each of the one or more vents is smaller than the treat access hole, wherein the annular side wall and the top wall intersect to form an annular rim about the circumference of the annular side wall, the annular rim sloping or curving axially inward and thereby defining an annular trough about the circumference of the annular side wall to redirect liquid on the top wall toward the interior portion of the pet toy; a core member configured to assemble to the bottom end of the cylindrical main body member, and comprising: a bottom end that is closed at a base; and an annular side wall, comprising mating features on a side surface thereof that are dimensioned to mate with the mating features of the cylindrical main body member

when the core member is assembled to the bottom end of the cylindrical main body member; and a treat container, disposed within an interior portion of the pet toy and configured to contain a pet treat, wherein the treat access hole faces toward the treat container when the core member is assembled to the cylindrical main body member, wherein a widest or largest dimension of the treat access hole is wide enough to allow passage or penetration of a tongue of a dog into the interior portion but smaller than a width of the treat container, wherein the cylindrical main body member is made from a pliable or flexible material that resists damage and permanent deformation, and wherein, when the core member is assembled to the cylindrical main body member, contact between the cylindrical main body member and the core member forms a liquid-tight or liquid-proof seal between the cylindrical main body member and the core member.

2. The pet toy of claim 1, wherein: the annular side wall projects axially outward in relation to the top wall where the annular side wall and the top wall intersect to form the annular rim.

3. The pet toy of claim 1, wherein the cylindrical main body member further comprises: a plurality of grooves disposed axially along the annular side wall.

4. The pet toy of claim 1, wherein: the treat container is generally bowl-shaped.

5. The pet toy of claim 1 wherein, as a result of the liquid-tight or liquid-proof seal between the cylindrical main body member and the core member, liquid flowing down the one or more vents toward the interior portion is prevented from contacting the mating features.

6. The pet toy of claim 1, wherein the base comprises: an annular flange that abuts the bottom end of the cylindrical main body member when the core member is inserted into the cylindrical main body member.

7. The pet toy of claim 1, wherein: the top wall of the cylindrical main body member is rounded axially outward; and the base of the core member is rounded axially outward.

8. The pet toy of claim 1, wherein: a center of gravity of the pet toy is disposed closer to the base than the top end of the cylindrical main body member to bias the top wall of the cylindrical main body member upward.

9. The pet toy of claim 1, wherein the core member comprises: a grip disposed on the base configured to be grasped by a human hand.

10. A kit comprising: the pet toy of claim 1; and the pet treat having a dimension larger than a dimension of the treat access hole so that the top wall of the cylindrical main body member traps the pet treat in the treat container of the core member.

11. A kit comprising: the pet toy of claim 1; and a mold configured to shape the pet treat having a dimension larger than a dimension of the treat access hole so that the top wall of the cylindrical main body member traps the pet treat in the treat container of the core member.

12. The pet toy of claim 1, wherein the treat container is radially defined and encircled by one or both of the annular side wall of the cylindrical main body member and the annular side wall of the core member.

13. The pet toy of claim 1, wherein the treat container comprises an inner side surface and an inner bottom surface.

14. The pet toy of claim 13, wherein the inner side surface and inner bottom surface are devoid of holes such that the treat container is configured to contain liquid.

15. The pet toy of claim 1, wherein the mating features of the cylindrical main body member and the mating features of the core member are configured to mate by screwing or twisting the core member into the bottom end of the cylindrical main body member.

16. The pet toy of claim 15, wherein the mating features of the cylindrical main body member and the mating features of the core member are threads.

17. A method for occupying a pet, comprising: loading a pet treat into the treat container of the pet toy of claim 1; and assembling the core member of the pet toy to the cylindrical main body member of the pet toy so that the treat access hole is facing toward the treat container, wherein the top wall of the cylindrical main body member traps the pet treat in the treat container while providing access

to the pet treat through the treat access hole.

18. The method of claim 17, wherein the assembling step comprises screwing or twisting the core member into the bottom end of the cylindrical main body member to cause the mating features of the cylindrical main body member and the mating features of the core member to mate.

19. The method of claim 18, wherein the mating features of the cylindrical main body member and the mating features of the core member are threads.

20. The pet toy of claim 1, wherein the treat access hole of the top wall of the cylindrical main body member is the only treat access hole of the cylindrical main body member and the core member.

21. The pet toy of claim 1, wherein the treat access hole of the top wall intersects a centerline axis of the pet toy.
